CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

Shortest Distances

Mark scheme 8 questions · 99 marks

Questions

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	Total	**99**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$\frac{\lvert 3(5)+1(-5)+2(3)+4\rvert}{\sqrt{3^2+1^2+2^2}}$	**M1**	Substitutes P and the plane into the distance formula, or finds the foot of the perpendicular. Condone one sign slip.
	$= \frac{20}{\sqrt{14}}$ $\left[= \frac{10\sqrt{14}}{7}\right]$	**A1**	$\frac{20}{\sqrt{14}}$ oe, or awrt 5.35.
		[2]	
1(b)	$2+\lambda = 5 \Rightarrow \lambda = 3$	**M1**	Equates one component of the general point of L to that of P and solves for λ.
	$y = 1-2(3) = -5$, $z = -3+2(3) = 3$ so P lies on L	**A1 AG**	Both other components shown to match P at $\lambda = 3$, or $\lambda = 3$ from all three, with a conclusion.
		[2]	
1(c)	$3(2+\lambda)+(1-2\lambda)+2(-3+2\lambda) = -4$ $\Rightarrow \lambda = \ldots$	**M1**	Substitutes all three components of L into the plane equation and solves for λ. Condone sign slips.
	$\lambda =$ "-1" $\Rightarrow (2-1,\ 1+2,\ -3-2)$	**dM1**	Substitutes their λ into the equation of L.
	$(1, 3, -5)$	**A1**	
		[3]	
1(d)	$\mathbf{u}\cdot\mathbf{n} = 3-2+4 = 5$, $\lvert\mathbf{u}\rvert = 3$, $\lvert\mathbf{n}\rvert = \sqrt{14}$	**B1**	All three values, seen or used.
	$\sin\theta = \frac{\text{"5"}}{\text{"3"}\ \text{"}\sqrt{14}\text{"}} \Rightarrow \theta = \ldots$	**M1**	Uses $\sin\theta = \frac{\lvert\mathbf{u}\cdot\mathbf{n}\rvert}{\lvert\mathbf{u}\rvert\lvert\mathbf{n}\rvert}$ with their values, or finds the angle with the normal by cosine and subtracts it from $90°$.
	$\theta = 26.5°$	**A1**	awrt $26.5°$, or awrt 0.462 radians.
		[3]	
1(e)	$PC = (3-(-1))\times 3 = \ldots$ $[= 12]$	**M1**	Finds PC as the change in λ times their $\lvert\mathbf{u}\rvert$, or as the magnitude of $\overrightarrow{PC}$ using their point from (c).
	"12" $\times$ "$\frac{5}{3\sqrt{14}}$"	**dM1**	Multiplies their PC by their $\sin\theta$ from (d), or by the cosine of their angle with the normal.

Question 1 continued

Question	Scheme	Marks	Notes
	$= \frac{20}{\sqrt{14}}$, as in (a)	**A1**	$\frac{20}{\sqrt{14}}$ oe, or awrt 5.35, with a statement that it agrees with (a).
		[3]	
	Total for Question 1	**13**	

Question	Scheme	Marks	Notes
2(a)	$\overrightarrow{CD} = \begin{pmatrix} 3 \\ 3 \\ -3 \end{pmatrix}$	**B1**	Or any nonzero multiple, such as $\mathbf{i}+\mathbf{j}-\mathbf{k}$.
	$\sin\alpha = \frac{\lvert(1)(1)+(1)(1)+(-1)(1)\rvert}{\sqrt{3}\sqrt{3}}$	**M1**	Uses the scalar product of their direction of CD and $\mathbf{i}+\mathbf{j}+\mathbf{k}$ over the product of magnitudes, as $\sin\alpha$, or as a cosine then subtracts from $90°$.
	$\sin\alpha = \frac{1}{3}$	**A1**	Or the cosine of the angle with the normal $= \frac{1}{3}$.
	$\alpha = 19.5°$	**A1**	
		[4]	
2(b)	$\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} + t\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$	**M1**	Uses C or D and their direction of CD from (a) in a vector or parametric equation.
	$(x, y, z) = (t, t, 2-t)$	**A1**	oe for their point and parameter.
	P_1: $t+t+2-t=3 \Rightarrow t=\ldots$ P_2: $2t-t=1 \Rightarrow t=\ldots$	**M1**	Substitutes their general point into both plane equations and solves each, or into one and tests the resulting point in the other.
	$t=1$ from both, so the point is common to P_1 and P_2	**A1**	The same parameter from both, or the point from one shown to satisfy the other, with a conclusion that it is common. A0 if the conclusion is missing.
	$(1, 1, 1)$	**A1**	
		[5]	
2(c)(i)	$(\mathbf{i}+\mathbf{j}+\mathbf{k}) \times (2\mathbf{i}-\mathbf{j})$ $= \mathbf{i}(0+1) - \mathbf{j}(0-2) + \mathbf{k}(-1-2)$	**M1**	Expands a determinant or uses the component formula for all three components. Condone one sign slip.
	$= \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$	**A1**	
		[2]	
2(c)(ii)	$\lvert\mathbf{i}+2\mathbf{j}-3\mathbf{k}\rvert = \sqrt{14}$ $\lvert\mathbf{i}+\mathbf{j}+\mathbf{k}\rvert = \sqrt{3},\ \lvert2\mathbf{i}-\mathbf{j}\rvert = \sqrt{5}$	**B1**	All three needed.
	$\sin\theta = \frac{\text{"}\sqrt{14}\text{"}}{\sqrt{3}\sqrt{5}} \Rightarrow \theta = \ldots$	**M1**	Divides the magnitude of their vector from (c)(i) by the product of the normals' magnitudes as $\sin\theta$, and finds θ. Not for a scalar-product route that doesn't use (c)(i).

Question 2 continued

Question	Scheme	Marks	Notes
	$\theta = 75.0°$	**A1**	
		[3]	
2(c)(iii)	$\begin{pmatrix}0\\0\\2\end{pmatrix} - \begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}-1\\-1\\1\end{pmatrix}$ $\begin{pmatrix}-1\\-1\\1\end{pmatrix} \times \begin{pmatrix}1\\2\\-3\end{pmatrix} = \ldots$	**M1**	Forms the vector from a point on both planes, such as their point from (b), to C, and crosses it with their direction from (c)(i). Condone sign slips.
	$\begin{pmatrix}1\\-2\\-1\end{pmatrix}$	**A1**	Or its negative.
	dist $= \frac{"\sqrt{6}"}{\sqrt{14}}$	**dM1**	Divides the magnitude of their cross product by the magnitude of their direction.
	$= \sqrt{\frac{3}{7}} \approx 0.655$	**A1**	
		[4]	
	Total for Question 2	**18**	

Question	Scheme	Marks	Notes
3(a)	$(1)(1)+(-2)(2)+(3)(1)=\ldots$ $(1)(4)+(-2)(-1)+(3)(-2)=\ldots$	**M1**	Forms the scalar product of the vector with both direction vectors, or finds the vector product of the direction vectors.
	$1-4+3=0$ and $4+2-6=0$ Perpendicular to both direction vectors $\Rightarrow$ perpendicular to Π	**A1 AG**	Both products shown evaluated to zero, or the vector product shown to be a multiple of the vector, with a conclusion.
		[2]	
3(b)	$x-2y+3z=(1)(2)+(-2)(-1)+(3)(3)=\ldots$	**M1**	Substitutes $(2,-1,3)$, or another point of Π, into $x-2y+3z=d$, or expands $(\mathbf{i}-2\mathbf{j}+3\mathbf{k})\cdot(\mathbf{r}-\mathbf{a})=0$. Condone sign slips.
	$x-2y+3z=13$	**A1**	Cartesian form, oe. $\mathbf{r}\cdot(\mathbf{i}-2\mathbf{j}+3\mathbf{k})=13$ alone scores A0.
		[2]	
3(c)	$A=(-2+t,\ 1+t,\ 4+t)$ $\frac{\lvert(-2+t)-2(1+t)+3(4+t)-\text{"}13\text{"}\rvert}{\sqrt{1+4+9}}$ $=\frac{9}{\sqrt{14}}$	**M1**	Uses a general point of ℓ in a point-to-plane distance with their plane from (b), set equal to $\frac{9}{\sqrt{14}}$, or an equivalent method such as parallel planes. Condone sign slips.
	$\lvert 2t-5\rvert=9$	**A1**	oe, such as $(2t-5)^2=81$ or $2t-5=\pm 9$.
	$2t-5=9$ or $2t-5=-9\Rightarrow t=\ldots$ $[=7,-2]$	**dM1**	Solves both cases, or their quadratic, for two values of t.
	$A=(5,8,11)$ or $(-4,-1,2)$	**A1**	Both points, as coordinates or position vectors, and no others.
		[4]	
	Total for Question 3	**8**	

Question	Scheme	Marks	Notes
4(a)	$\overrightarrow{AB} = \begin{pmatrix}1\\2\\0\end{pmatrix}$, $\overrightarrow{AC} = \begin{pmatrix}1\\0\\-1\end{pmatrix}$ $\overrightarrow{AB} \times \overrightarrow{AC} = \ldots$	**M1**	Finds two non-parallel vectors in the plane and attempts their cross product, or solves two scalar products with them equal to zero. Condone sign slips.
	$\mathbf{n} = \begin{pmatrix}-2\\1\\-2\end{pmatrix}$	**A1**	Any nonzero multiple.
	$2x - y + 2z = 2(1) - (-2) + 2(3)$	**M1**	Substitutes a point of Π into $ax + by + cz = d$ or $\mathbf{r} \cdot \mathbf{n} = d$, with their normal, to find d.
	$2x - y + 2z = 10$	**A1**	oe, including $\mathbf{r} \cdot \begin{pmatrix}2\\-1\\2\end{pmatrix} = 10$ or any multiple.
	$\text{dist} = \frac{\lvert 2(4)-1+2(0)-10\rvert}{\sqrt{2^2+(-1)^2+2^2}} = \ldots$	**M1**	Uses the point-to-plane formula with D and their plane $ax + by + cz = d$, or projects $\overrightarrow{AD}$ onto their normal. Condone sign slips.
	$= 1$	**A1**	
		[6]	
4(b)	$\begin{pmatrix}2\\1\\2\end{pmatrix} \times \begin{pmatrix}1\\-2\\1\end{pmatrix} = \ldots$	**M1**	Attempts the cross product of the two direction vectors, or solves two scalar products with them equal to zero. Condone sign slips.
	$= \begin{pmatrix}5\\0\\-5\end{pmatrix}$	**A1**	Any nonzero multiple.
	$\begin{pmatrix}2\\-3\\5\end{pmatrix} \cdot$ " $\begin{pmatrix}5\\0\\-5\end{pmatrix}$ " $[= -15]$ $\text{dist} = \frac{15}{\sqrt{5^2+0^2+(-5)^2}} = \ldots$	**dM1**	Divides the scalar product of a vector joining the lines with their normal by the normal's magnitude. Condone a missing modulus.
	$= \frac{3\sqrt{2}}{2}$	**A1**	Any exact form, such as $\frac{3}{\sqrt{2}}$.
		[4]	
	Total for Question 4	**10**	

Question	Scheme	Marks	Notes
5(i)	Sets $z = t$: $x - y + t = 6,\ 2x + y - t = 3$ $\Rightarrow x = \ldots,\ y = \ldots$	**M1**	Solves the plane equations with one variable as parameter, or crosses the normals and finds a common point. Condone slips.
	$\mathbf{r} = \begin{pmatrix} 3 \\ -3 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$	**A1**	Any point on ℓ_2 and any multiple of the direction.
	$\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \times " \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} " = \ldots$	**M1**	Forms the vector product of the two directions, or finds a vector perpendicular to both. At least two components correct for their vectors.
	$= \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$	**A1**	Or any nonzero multiple.
	$\dfrac{\left\lvert \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix} \cdot " \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} " \right\rvert}{\sqrt{11}} = \ldots$	**dM1**	Divides the scalar product of a vector joining the two lines with their normal by the normal's magnitude. Condone a missing modulus.
	$\dfrac{2}{\sqrt{11}}$	**A1**	oe, or awrt 0.603.
		[6]	
5(ii)	$" \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} " \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = \ldots$	**M1**	Forms $ax + by + cz = d$ with their normal from (i) and finds d from a point on ℓ_1.
	$3x - y + z = 10$	**A1**	Any Cartesian equivalent. Not for a vector equation alone.
		[2]	
	Total for Question 5	**8**	

Question	Scheme	Marks	Notes
6(a)	$\det\begin{pmatrix}1 & \lambda & 1\\ 2 & -1 & 2\\ 2 & 1 & -1\end{pmatrix}$ $= 1(1-2) - \lambda(-2-4) + 1(2+2)$	**M1**	Expands the determinant of the coefficient matrix by any row or column, giving an expression in λ. Condone sign slips.
	$= 6\lambda + 3$	**A1**	oe, such as $3(2\lambda+1)$.
	$6\lambda + 3 = 0 \Rightarrow \lambda = -\frac{1}{2}$ Unique point for all $\lambda \neq -\frac{1}{2}$	**A1**	Needs the conclusion that the point is unique for every other λ. The case $\lambda = -\frac{1}{2}$ need not be examined further.
		[3]	
6(b)	$\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}1 & 2 & 1\\ 2 & -1 & 2\\ 2 & 1 & -1\end{pmatrix}^{-1}\begin{pmatrix}4\\3\\5\end{pmatrix}$	**M1**	Writes the system as $\mathbf{Ax} = \mathbf{b}$ with $\lambda = 2$ and uses $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$. Not for solving the equations without matrices.
	$\mathbf{A}^{-1} = \frac{1}{15}\begin{pmatrix}-1 & 3 & 5\\ 6 & -3 & 0\\ 4 & 3 & -5\end{pmatrix}$	**B1**	Any equivalent form; an inverse from a calculator is accepted.
	$\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}2\\1\\0\end{pmatrix}$, $P = (2,1,0)$	**A1**	
		[3]	
6(c)	$\frac{x+1}{1} = \frac{y-1}{-1} = \frac{z}{1} = t$ $\Rightarrow x = t-1,\ y = 1-t,\ z = t$	**M1**	Sets the ratios equal to a parameter and writes x, y, z in terms of it, or reads a point and a direction from the Cartesian form. Condone sign slips.
	$\mathbf{r} = \begin{pmatrix}-1\\1\\0\end{pmatrix} + t\begin{pmatrix}1\\-1\\1\end{pmatrix}$	**A1**	Any point of ℓ and any multiple of its direction, with $\mathbf{r} =$ or equivalent.
		[2]	
6(d)	$"\begin{pmatrix}2\\1\\0\end{pmatrix}" - \begin{pmatrix}-1\\1\\0\end{pmatrix} = \begin{pmatrix}3\\0\\0\end{pmatrix}$ $\begin{pmatrix}3\\0\\0\end{pmatrix} \times \begin{pmatrix}1\\-1\\1\end{pmatrix} = \ldots$	**M1**	Forms the vector from any point of ℓ to their P and its vector product with the direction of ℓ. Condone sign slips.
	$= \begin{pmatrix}0\\-3\\-3\end{pmatrix}$	**A1**	oe, such as its negative, or the multiple that their multiple of the direction gives.

Question 6 continued

Question	Scheme	Marks	Notes
	$\dfrac{\sqrt{0^2+(-3)^2+(-3)^2}}{\sqrt{1^2+(-1)^2+1^2}} = \ldots$	**dM1**	Divides the magnitude of their vector product by the magnitude of their direction of ℓ.
	$= \sqrt{6}$	**A1**	$\sqrt{6}$ or awrt 2.45.
		[4]	
6(e)(i)	$\begin{pmatrix}1\\-1\\1\end{pmatrix} \cdot \begin{pmatrix}2\\1\\-1\end{pmatrix} = 2-1-1$	**M1**	Forms the scalar product of the direction of ℓ with the normal of the plane.
	$= 0$	**A1**	
	$2(-1)+1-0 = -1 \neq 5$ so ℓ is not in the plane, and is parallel to it	**A1 AG**	Needs the zero scalar product and a point of ℓ shown not to satisfy the plane's equation, then a conclusion of parallel in words.
		[3]	
6(e)(ii)	$\dfrac{\lvert 2(-1)+1(1)-1(0)-5\rvert}{\sqrt{2^2+1^2+(-1)^2}} = \ldots$	**M1**	Substitutes any point of ℓ into the point-to-plane distance formula, or uses the parallel plane through ℓ. Unquoted, implied only by a correct substitution.
	$= \frac{6}{\sqrt{6}} = \sqrt{6}$	**A1**	$\sqrt{6}$ or awrt 2.45.
		[2]	
	Total for Question 6	**17**	

Question	Scheme	Marks	Notes
7(a)(i)	$\mathbf{u}\cdot\mathbf{v}=\lambda-4$	**B1**	
		[1]	
7(a)(ii)	$\mathbf{u}\times\mathbf{v}=(2-2)\mathbf{i}-(-\lambda-2)\mathbf{j}$ $+(\lambda+2)\mathbf{k}$	**M1**	Attempts all three components by a determinant or the component formula. Condone sign slips.
	$=(\lambda+2)(\mathbf{j}+\mathbf{k})$	**A1**	oe, such as a column vector; need not be factorised.
		[2]	
7(b)(i)	$\lvert 2\mathbf{i}-2\mathbf{j}+2\mathbf{k}\rvert=2\sqrt{3}$, $\lvert\mathbf{i}+\mathbf{j}-\mathbf{k}\rvert=\sqrt{3}$	**B1**	
	$\cos\theta=\dfrac{\lvert"2-4"\rvert}{2\sqrt{3}\times\sqrt{3}}=\ldots$	**M1**	Divides their (a)(i) at $\lambda=2$ by the product of their magnitudes in the cosine formula. Condone a missing modulus.
	$\theta=70.5°$	**A1**	awrt $70.5°$ or 1.23 rad, or $\cos^{-1}\frac{1}{3}$.
		[3]	
7(b)(ii)	$(\mathbf{i}-2\mathbf{j}+2\mathbf{k})\times(\mathbf{i}+\mathbf{j}-\mathbf{k})=3(\mathbf{j}+\mathbf{k})$	**B1ft**	ft their (a)(ii) at $\lambda=1$; any nonzero multiple accepted.
	$\dfrac{\lvert(\mathbf{i}+3\mathbf{j}+4\mathbf{k})\cdot"(3\mathbf{j}+3\mathbf{k})"\rvert}{\lvert"3\mathbf{j}+3\mathbf{k}"\rvert}$ $=\ldots$	**M1**	Divides the scalar product of a vector joining the two lines with their common normal by its magnitude. Condone sign slips in the joining vector.
	$\dfrac{7\sqrt{2}}{2}$	**A1**	$\frac{7\sqrt{2}}{2}$ or $\frac{7}{2}\sqrt{2}$. Not for $\frac{7}{\sqrt{2}}$ or a decimal alone.
		[3]	
	Total for Question 7	**9**	

Question	Scheme	Marks	Notes
8(a)	$\overrightarrow{AD} = \begin{pmatrix}1\\1\\m\end{pmatrix}$, $\overrightarrow{BC} = \begin{pmatrix}-1\\1\\2\end{pmatrix}$, $\overrightarrow{AB} = \begin{pmatrix}1\\0\\0\end{pmatrix}$	**B1**	All three, oe; any vector joining a point of one line to a point of the other in place of $\overrightarrow{AB}$.
	$\begin{pmatrix}1\\1\\m\end{pmatrix} \times \begin{pmatrix}-1\\1\\2\end{pmatrix} = \begin{pmatrix}2-m\\-2-m\\2\end{pmatrix}$	**M1 A1**	**M1** Attempts the vector product of their two direction vectors. Condone sign slips. **A1** oe, such as its negative.
	$\dfrac{\left\lvert\overrightarrow{AB} \cdot \text{"}\mathbf{n}\text{"}\right\rvert}{\lvert\text{"}\mathbf{n}\text{"}\rvert} = \dfrac{1}{\sqrt{14}}$, $\mathbf{n}$ their vector product	**M1**	Uses the scalar product of their joining vector with their vector product, divided by its magnitude, and sets it equal to $\frac{1}{\sqrt{14}}$.
	$\dfrac{\lvert 2-m\rvert}{\sqrt{2m^2+12}} = \dfrac{1}{\sqrt{14}}$	**A1**	Correct equation oe, need not be simplified. Condone a missing modulus.
	$14(2-m)^2 = 2m^2+12$ $\Rightarrow 3m^2 - 14m + 11 = 0 \Rightarrow m = \ldots$	**dM1**	Squares and solves their three-term quadratic in m by any means, a calculator included.
	$m = 1$ or $m = \frac{11}{3}$; m is an integer, so $m = 1$	**A1 AG**	cso. Both roots from a correct quadratic, with $\frac{11}{3}$ rejected as not an integer. A0 if $m = 1$ is only verified.
		[7]	
8(b)	$\overrightarrow{AC} \times \overrightarrow{AD} = \begin{pmatrix}0\\1\\2\end{pmatrix} \times \begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}-1\\2\\-1\end{pmatrix}$	**M1 A1**	**M1** Attempts the vector product of a vector from a point on the line to C with the line's direction, using $m = 1$. Condone sign slips. **A1** Or its negative.
	$d = \dfrac{\lvert\text{"}-\mathbf{i}+2\mathbf{j}-\mathbf{k}\text{"}\rvert}{\left\lvert\overrightarrow{AD}\right\rvert} = \dfrac{\sqrt{6}}{\sqrt{3}}$	**dM1**	Divides the magnitude of their vector product by the magnitude of their direction vector.
	$d = \sqrt{2}$	**A1**	Or awrt 1.41.
		[4]	
8(c)	$\overrightarrow{AB} \times \overrightarrow{AD} = \begin{pmatrix}1\\0\\0\end{pmatrix} \times \begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}0\\-1\\1\end{pmatrix}$	**M1 A1**	**M1** Attempts the vector product of two non-parallel vectors in plane ABD, or finds a normal from two scalar products set to zero. Condone sign slips. **A1** Any nonzero multiple.

Question 8 continued

Question	Scheme	Marks	Notes
	$\mathbf{n}_{ACD} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$	**B1**	Any nonzero multiple; may be taken from (b).
	$\cos\theta = \dfrac{\lvert\text{"}(0,-1,1)\text{"} \cdot \text{"}(-1,2,-1)\text{"}\rvert}{\sqrt{2}\sqrt{6}}$	**M1**	Uses the scalar product of their two normals divided by the product of their magnitudes.
	$\cos\theta = \dfrac{3}{\sqrt{12}} = \dfrac{\sqrt{3}}{2}$ so the acute angle is $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$	**A1 AG**	cso, from correct normals. A negative cosine must be made positive, or the obtuse angle replaced by its supplement. Needs a concluding statement.
		[5]	
Total for Question 8		**16**	